

- 1 The bacterium *Escherichia coli* uses glucose as the main respiratory substrate, but other substances, such as lactose, can be used when necessary.

The enzyme β -galactosidase hydrolyses lactose to form glucose and galactose, which can then be used by *E. coli*.

Production of β -galactosidase in *E. coli* is regulated and involves a length of DNA known as the *lac* operon.

- (a) Figure 1.1 is a simple diagram of the *lac* operon.

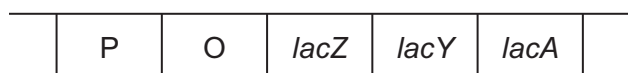


Figure 1.1

When lactose is absent, a repressor molecule is attached to part of the *lac* operon.

Draw a cross (X) on Figure 1.1 to show the location where the repressor attaches to the *lac* operon when lactose is absent. [1]

- (b) Structural genes *lacZ*, *lacY* and *lacA* are parts of the *lac* operon.

Complete Table 1.1 to:

- name the gene product of *lacY* and of *lacA*
- state the role of the gene product of *lacY*.

Table 1.1

structural gene	gene product	role of gene product
<i>lacZ</i>	β -galactosidase	hydrolyses lactose to form glucose and galactose
<i>lacY</i>		
<i>lacA</i>		transfers an acetyl group from one molecule to another

[2]

- (c) The enzyme β -galactosidase is an inducible enzyme.

Explain what is meant by an inducible enzyme.

.....

..... [1]

- A sample of *E. coli* was removed from a liquid nutrient medium containing glucose.
- The sample was added to a liquid nutrient medium containing limited quantities of glucose and lactose.
- Estimates of the total number of cells in the sample (population size) were made at regular intervals.

The graph shows the growth of *E. coli* population size over time, divided into three phases: A, B, and C. The y-axis represents population size, and the x-axis represents time. Phase A shows a constant population size. Phase B shows a linear increase in population size. Phase C shows a constant population size.

Phase	Population Size Change
A	Constant
B	Linear Increase
C	Constant

With reference to Figure 1.2, suggest explanations for phase **B** and phase **C**.

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[4]

[Total: 8]